Faculty of Computer Science

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LawVoice: A Voice-Based Intelligent Assistant for Exploring Legal Scenarios and Developing Critical Thinking in Teenagers

Student: ____________________________
Academic Supervisor:
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Current MVP: allaw-urist.ru

Contents

Topic

LawVoice is a voice-first educational assistant for helping teenagers explore legal situations.

Current state

Current MVP interface, LawVoice personas, dialogue logic, knowledge base, voice flow, and action-plan prototype.

Still in progress

Student name, supervisor, pilot-study results, evaluation metrics, and the final expanded bibliography.

Subject Area

This work focuses on the design and current MVP implementation of LawVoice, a voice-based educational assistant that helps teenagers discuss everyday legal situations in a safe format.

- The project is positioned as an educational voice simulator, not as a legal consultation service.
- The current MVP centers on cyberbullying, online purchases, and rights at school.
- The implemented dialogue flow clarifies facts, explains risks, suggests options, and ends with a concrete next step.
- The LawVoice layer is built on top of an adapted OpenAI realtime voice-agent base.

Core idea

The aim is not simply to give an answer, but to guide the user through a structured chain: facts -> risks -> options -> next step.

Interaction format

Voice-first - educational - scenario-based

Research Relevance

- LawVoice addresses everyday situations that teenagers may face outside a formal law class.
- Voice interaction lowers the entry barrier because speaking through a case can be easier than reading long legal texts.
- The educational value is connected not only with legal literacy, but also with critical thinking, because the user compares options and consequences.
- The project combines AI dialogue, safe communication rules, and a practical learning interface in one MVP.

MVP scenarios

Cyberbullying

Online purchases

Rights at school

Goal and Objectives

Goal

To develop and present the current MVP of a voice-based intelligent assistant that helps teenagers explore legal situations and practice structured reasoning.

Objectives

1. Analyze teen legal scenarios and define a safe educational framing.

2. Design dialogue logic, personas, age adaptation, and escalation rules.

3. Implement the current MVP interface, voice interaction, and knowledge-base-supported action planning.

4. Define limitations and next steps for legal, pedagogical, and empirical evaluation.

Research Roadmap

1. Subject area analysis
2. Scenario and role design
3. Dialogue logic development
4. MVP interface implementatio n
5. Testing and learning analytics
6. Impact evaluation and final reporting

At the current project stage, steps 1-4 are represented in the MVP. Steps 5-6 remain the next research and deployment stage.

User Journey

1. Case selection

The teenager selects a situation and one of the LawVoice personas.

2. Fact clarification

The assistant asks short clarifying questions without requesting unnecessary personal data.

3. Analysis

LawVoice explains rights, obligations, and risks in plain language.

4. Action options

The user receives 2-3 possible legal and safe next steps.

5. Next step

The dialogue ends with a recommended action and a short 24-hour checklist.

Educational outcome: the user practices structured reasoning instead of waiting for a ready-made verdict.

Current MVP Functionality

Persona profiles
Mentor, peer, analyst, and safety personas with voice and style settings.

Adaptive mode
Teen and 18+ modes with runtime adjustment for risk and anxiety.

Base prompt
LawVoice system prompt and dialogue rules at session start.

Scenario and materials
Scenario hints, supporting materials, and prepared context for the conversation.

Knowledge base
Document upload, chunking, search, and token estimates.

Voice dialogue
Realtime session, microphone, audio visualization, and context prompts.

Action plan
Grounded draft plan with steps, priorities, and comments.

Analytics and tech panel
Session log, message count, token usage, and API cost.

Solution Architecture

Interface layer

- profile selection and dialogue-style screen
- age mode and prompt editing
- scenario and document input
- voice-session controls
- audio, conversation, and log widgets

Logic layer

- intent routing and LawVoice dialog manager
- session state and safety-aware profile switching
- grounded action-plan generation
- analytics logging and prompt control

AI and integration layer

- OpenAI Realtime API and WebRTC
- Node.js / Express backend
- PostgreSQL with pgvector and knowledge chunks
- gpt-realtime-mini and embedding services

Current implementation: a LawVoice domain layer built on top of an adapted realtime voice-agent foundation.

Teen Legal Scenarios

Cyberbullying

- clarify what happened, where it happened, and whether evidence exists
- separate facts from emotion and assess risk
- suggest safe next steps and evidence preservation
- in dangerous situations, recommend a trusted adult or 112

Online purchases

- clarify the order, payment, delivery, and seller
- review evidence such as receipts, chats, screenshots, and order cards
- discuss return, complaint, and documentation of the problem
- offer a practical next step for today

Rights at school

- discuss the conflict calmly and without stigma
- explain rights, obligations, and school boundaries in simple language
- help formulate a message for a teacher, parent, or administrator
- turn the situation into a short action plan

Interface Components

Session setup

Profile selection
Age mode
Rules and instructions
Scenario and document input

Voice dialogue

Start and end session
Microphone
Audio input and output
Urgent and quiet prompts

Reflection and control

Draft action plan
Learning analytics
Action log
Technical panel and cost

This layout separates preparation, live dialogue, and post-session reflection, which is useful both for a demo and for later research.

Current Limitations

- The current version is an MVP, and some interface and logic blocks are still demonstrational rather than research-validated.
- There are no completed empirical results or classroom pilot measurements yet.
- The legal content is currently tailored to Russian-language scenarios and still needs expert legal and pedagogical review.
- Educational deployment would require clearer privacy, moderation, and escalation procedures.

Important

The presentation intentionally avoids unsupported claims about learning impact or legal accuracy.

Future Development

Content

Expansion of scenarios, documents, explanation modes, and role profiles.

Safety

Refinement of escalation rules, moderation logic, and privacy boundaries.

Evaluation

Pilot testing and measurement of legal-literacy and critical-thinking outcomes.

- Add new scenarios, including peer-pressure and detention-related support.
- Replace remaining generic elements with fully LawVoice-specific components.
- Prepare pilot testing in an educational environment.
- Develop metrics for response quality, safety, and educational usefulness.

Expected Results and Conclusions

1. The project defines a feasible concept of a voice-first educational legal assistant for teenagers.
2. The current MVP shows that realtime voice interaction can support structured reasoning: facts, risks, options, and the next step.
3. The prototype provides a basis for future expert review, pilot testing, and impact evaluation rather than finalized legal guidance.

References

1. LawVoice MVP. https://allaw-urist.ru/

2. OpenAI. Realtime API documentation. https://platform.openai.com/docs/api-reference/realtime-sessions

3. LawVoice project repository. README and current MVP implementation.

4. Project note: LawVoice General Character Prompt.

5. Project note: Voice Assistant Architecture.

Appendix: Draft Main Screen Structure

Left column

Persona and dialogue style
Age mode
Rules
Base prompt
Profiles

Central area

Voice dialogue
Start / end session
Microphone / audio visualization

Right column

Scenario hints and materials
Knowledge base

Lower central area

Dialogue
Action log
Draft action plan

Lower right area

Learning analytics
Technical panel
API cost

Appendix: Educational Voice Session Logic

Situation
The user describes the case or chooses a scenario

→

Facts
The assistant clarifies key details

→

Rules and risks
LawVoice explains rights and safety factors in simple language

→

Decisions
2-3 possible next steps

→

Outcome
Recommended action + 24-hour checklist

This flow is suitable both for a product demonstration and for a later educational study.

Thank you for your attention